Week_7-8_Assignment

Velazquez-Feliciano, Adrian

DSC 640: Data Presentation and

Visualization 02/15/2026

Lottery Patterns, Randomness, and Responsible Community Pool

Audience:

My audience is adults in my community who already buy lottery tickets occasionally and are considering joining a small lottery pool. Most of them are not technical, so my goal is to explain probability concepts visually and plainly without implying guaranteed financial outcomes.

Purpose

The purpose of this presentation is to correct common misconceptions about lottery patterns. People often believe in hot numbers, due numbers, lucky weekdays, or that the previous draw influences the next one. Using historical draw data, the visuals show that outcomes behave like independent random draws. Any bumps or streaks that appear are the normal noise you expect from randomness, not signals you can exploit. After addressing the myths, I propose a responsible alternative for people who choose to play anyway: a transparent, budget-capped community lottery pool. Pooling does not improve the odds of any single ticket, but it does let a group buy more unique combinations for the same per-person cost. That increases coverage of the number space and reduces duplicate picks, which is the only practical advantage players can control.

Call to action: if you already plan to play, join the community pool today by committing a fixed weekly amount, agreeing to simple written rules for ticket selection and payout splitting, and participating in a transparent process where everyone can verify the tickets before the drawing.

Medium:

The medium is a short slide-style PDF designed for projection and group discussion. I use simple, familiar chart types so the audience can understand each figure quickly: bar charts, scatterplots, a histogram, a pie chart, and a line chart. I also keep chart titles short and neutral to avoid over-claiming.

Ethical Considerations:

This project avoids framing gambling as an investment. The visuals do not promise profit or predictive power. Simulation is labeled as a model for understanding coverage, not as proof of profitability. The overall message is pro-transparency and pro-responsibility: if people play, they should do so with clear rules, a spending cap, and honest expectations.

Visuals:

Below are the the visuals as pdf delivery for people to concider joining a pool for responsible and smart approach to lottery ticket purchases.

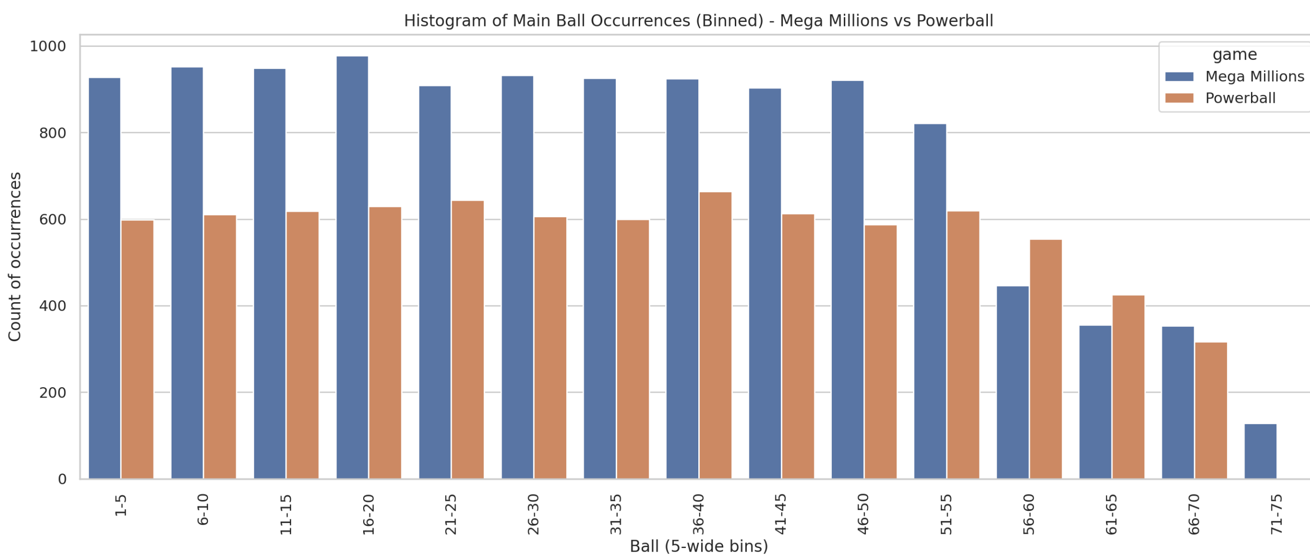
Visuals:

Number ranges look even (Mega Millions vs Powerball)

This histogram groups the main balls into small number ranges and counts how often outcomes fall into each range. If one part of the number line truly had an advantage, you would expect a consistent ‘favorite zone’ that towers above the rest. Instead, the bars are fairly balanced across bins for both Mega Millions and Powerball. Some bins are a little higher and some are a little lower, but that unevenness is exactly what randomness looks like when you summarize a large set of real draws.

A helpful way to picture this is to imagine two people in the room debating. One says, ‘Always pick low numbers,’ and the other says, ‘Always pick high numbers.’ This chart acts like a referee. If Team Low (say 1–25) truly had a built-in edge over Team High (say 50–75), it would show up as a persistent pattern in these binned counts. Because we do not see that kind of stable advantage, the takeaway is that range-based picking is not a reliable strategy.

This matters for the pool because it shifts our focus away from superstition and toward what we can actually control: coordination and coverage.

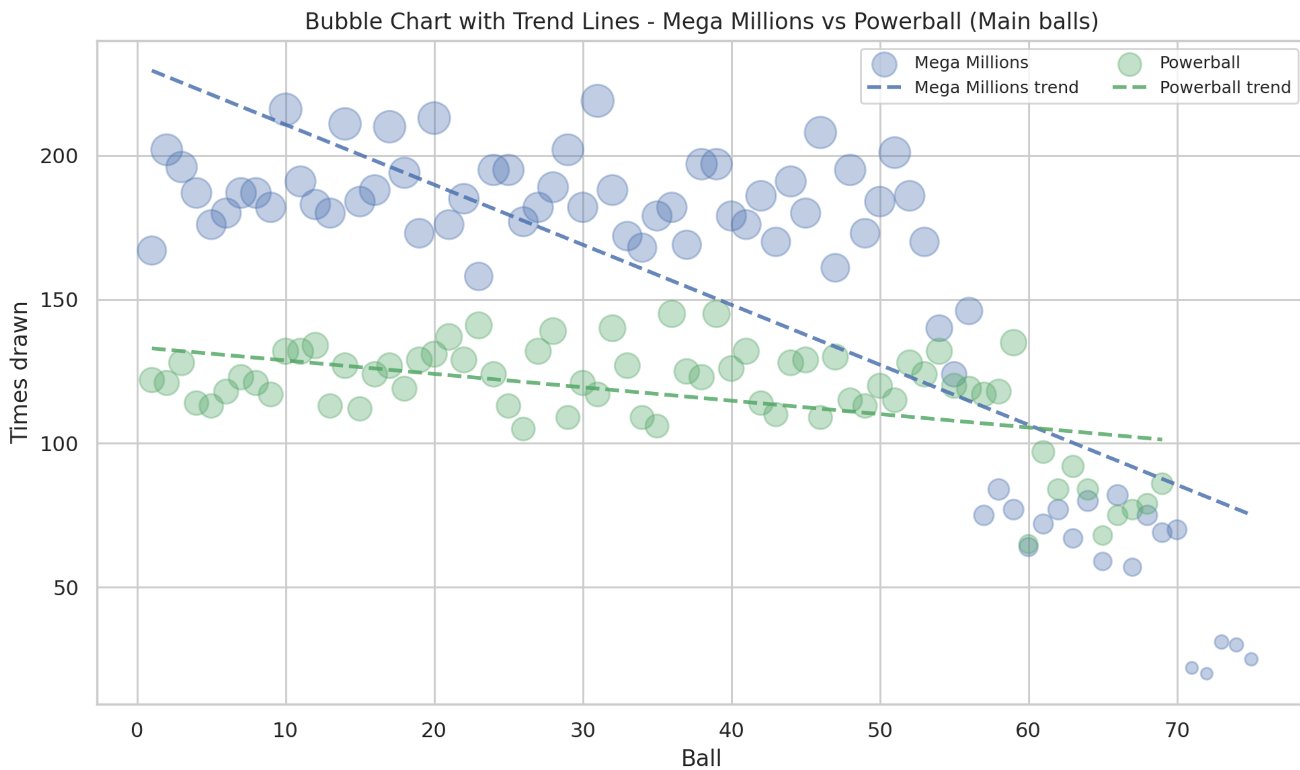


Ball frequency by number (Mega Millions vs Powerball)

This bubble plot is a close-up of the same idea, but at the level of individual numbers. Each dot represents one ball number and how many times it has been drawn historically. People naturally remember recent streaks and start to feel like certain numbers are ‘hot.’ But when you see the full set, the dots form a noisy band rather than a clear pattern that would justify chasing specific numbers.

If someone says, ‘My number 17 is hot—I keep seeing it,’ this visual is the calm response. Yes, some numbers will sit a bit above average and some a bit below average, and that can persist for a while simply by chance. The important point is that the overall trend is essentially flat: there is no stable, exploitable advantage tied to a particular number.

The pool lesson is straightforward. Since we cannot reliably pick ‘better’ numbers, our best move is not to overthink the selection based on perceived heat. Instead we use pool rules that emphasize unique combinations and avoid duplicates within the group.

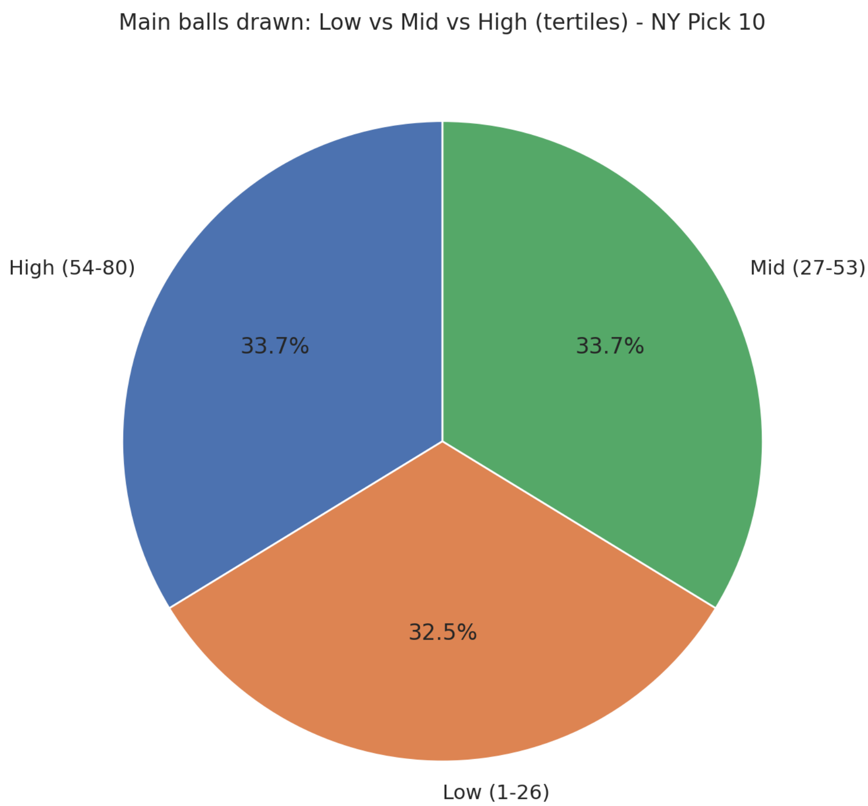


Low / mid / high numbers are balanced (NY Pick 10)

This pie chart compresses the number line into three big sections: low, mid, and high. It answers a very common belief: ‘Low numbers come out more.’ If that belief were true in a meaningful way, one slice would dominate. Instead, the slices are close to even. That is what you expect when the game is sampling fairly from the whole range over many draws.

A simple scenario is to think of the number range like three shelves in a store. If the lottery is fair, you should be ‘pulling items’ from the bottom shelf, middle shelf, and top shelf at roughly similar rates over time. This chart shows that’s what happens in the data.

For the audience, this is a useful ‘reset’ visual. It makes it easier to let go of low-number bias and accept that even an intuitive rule of thumb does not create an edge.

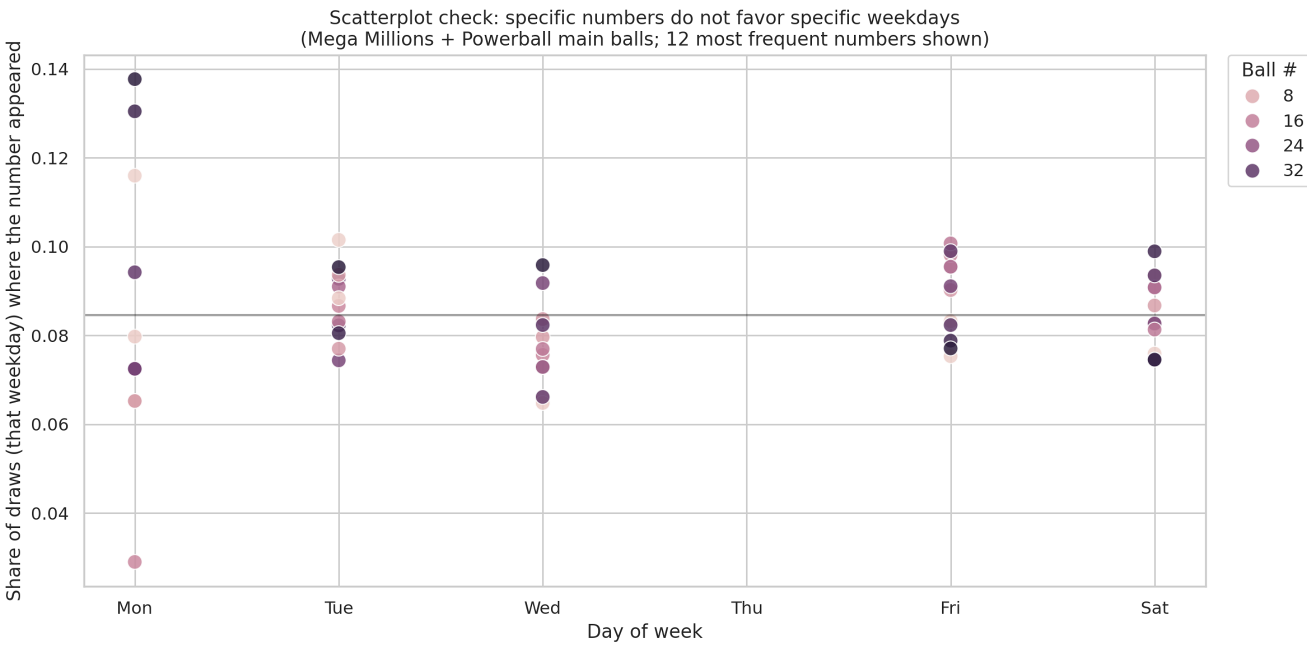


No special weekday for any number

This scatterplot tests the ‘lucky weekday’ idea. If certain numbers truly appeared more often on certain days, you would see consistent clusters or upward shifts for specific weekdays. Instead, the points form a broad cloud without a stable weekday signature.

In everyday terms, this is how coincidences fool us. People notice a number hit on a Friday a couple of times and build a story around it. The data, when you look at it all at once, does not support that story. The weekday does not offer a predictable advantage for choosing numbers.

For the pool, this reinforces a responsible message. We do not schedule buying or number selection around superstition. We keep the process simple, repeatable, and verifiable.

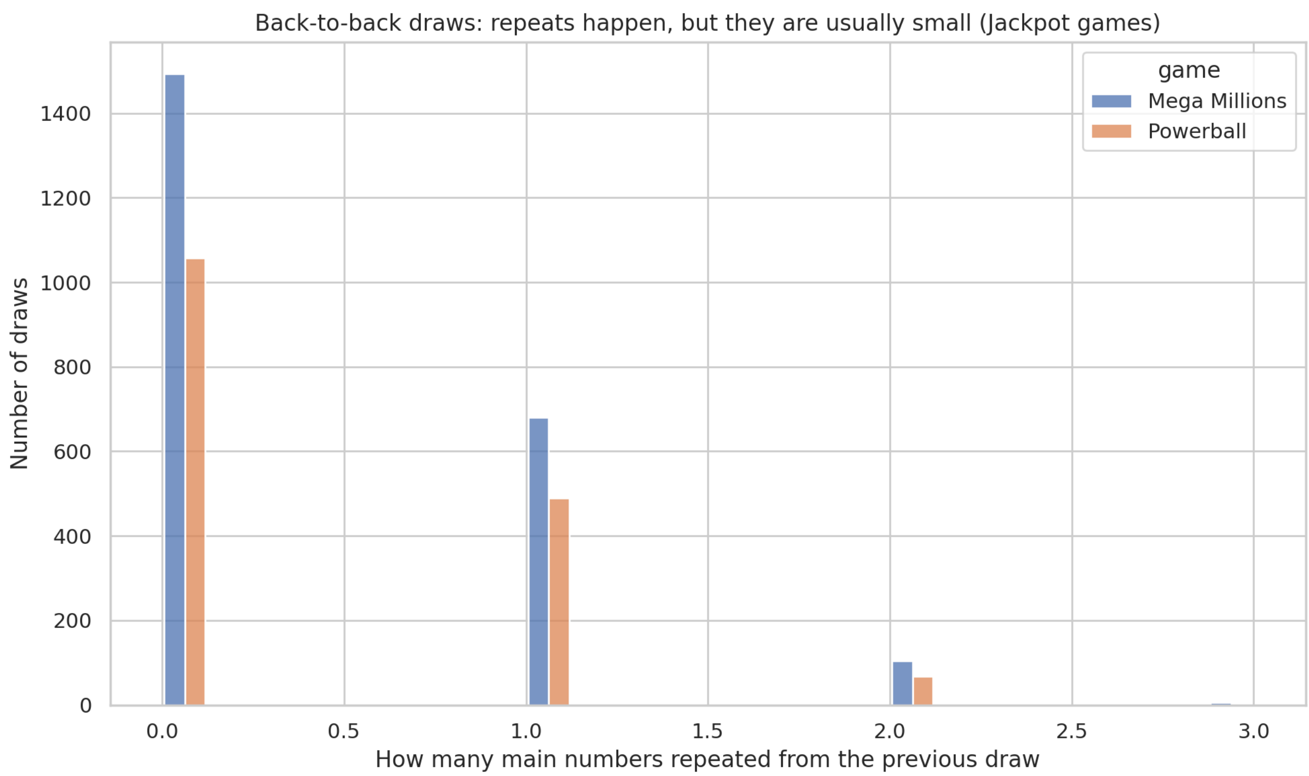


Small repeats between draws (Jackpot games)

This is a bar plot that looks at back-to-back draws and counts how many main numbers repeat from the previous draw. People often react strongly when they notice a repeat, and the immediate instinct is to treat it as a signal: 'That number is hot; it will show up again.' This visual puts repeats into context. Repeats do happen, but most of the time the number of repeats is small.

A simple scenario is to compare it to coin flips. Seeing heads twice in a row is real, and it feels meaningful, but it is also a normal outcome of randomness. Similarly, seeing a repeated main number from one jackpot draw to the next is not evidence that the lottery is 'stuck' on a number. It is simply one of the ordinary patterns you expect to see sometimes.

The pool takeaway is that we should not chase the last draw. Our rules should not give extra weight to numbers just because they appeared recently.

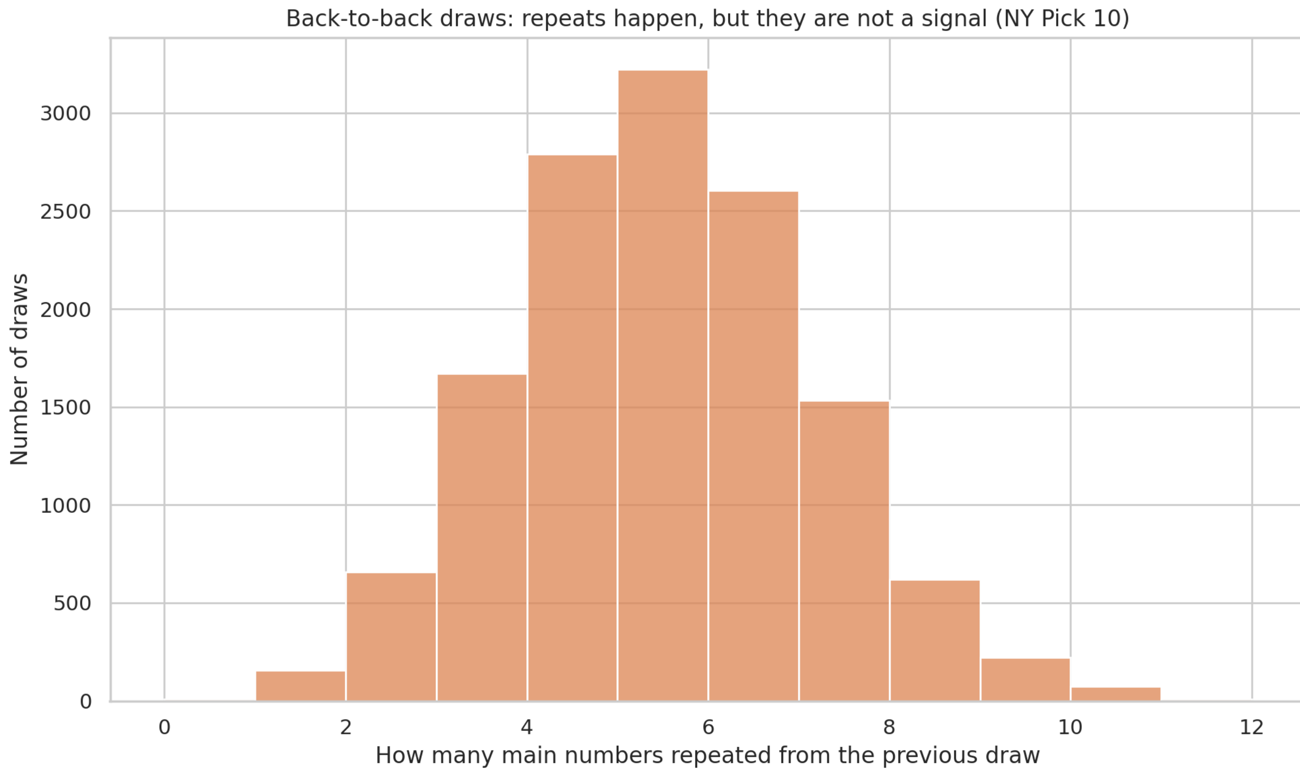


Repeat pattern between draws (NY Pick 10)

This repeat distribution makes the same point for a game that draws many numbers at once. In Pick 10, overlap between consecutive draws is naturally more common because each draw contains a larger set of numbers. That can make repeats feel suspicious or 'too frequent,' but the overall shape here is exactly what you expect from independent draws: most of the time you see a moderate overlap, with fewer cases of very low or very high overlap.

An easy analogy is to imagine two groups of 10 students chosen from the same school of 80. It would not be surprising if a few names show up in both groups. That overlap does not mean the second group copied the first; it is just how sampling works when your groups are large.

Again, the pool implication is about discipline. We do not treat overlap as a prediction tool. We focus on buying more unique combinations, not 'following' the previous draw.



More tickets means more number coverage (simulation)

This is the one visual that supports the practical community-pool idea. It shows a simulation-style relationship: as the number of tickets purchased increases, the share of the number space you cover increases too. Importantly, this is not a promise of profit and not a claim that the pool can predict anything. It is simply a statement about coverage. If you buy more unique combinations, you give yourself more ways to match the drawn numbers.

Scenario example: imagine 20 people who each spend the same small amount every week. If they all buy individually, some of them will accidentally choose the same combinations, and the group’s money will not be used efficiently. If those same 20 people pool their money and agree to avoid duplicates, the group buys a broader set of unique tickets. The odds per ticket do not change, but the group’s total coverage improves for the same per-person budget.

This is where the presentation ends with a responsible call to action. If you already plan to play, join the pool by committing a fixed weekly amount, agreeing to written rules for ticket selection and payout splitting, and using a transparent process where everyone can verify the tickets before the drawing.